

CSSCM 2026 COHORT

Ubuntu Python & NumPy Setup Guide

Step-by-Step Student Environment Preparation

Director	Elliot S. Menkah (PhD)
Tutors	Emmanuel Oppong (MPhil); Obed Adu-Gyamfi (BSc)

Goal: Prepare a clean Ubuntu environment in which you can create and run Python scripts using NumPy.

Open the Ubuntu Terminal before beginning. Commands shown below are entered into the terminal, not into Python itself.

1. Check Python 3

```
python3 --version
```

If a Python 3 version is displayed, continue. If the command is not found, install Python using Step 2.

2. Install Python 3 and Basic Tools (if needed)

```
sudo apt update
sudo apt install python3 python3-pip python3-venv
```

Enter your Ubuntu password if requested. The password is not displayed while you type; this is normal.

3. Check pip

```
python3 -m pip --version
```

This confirms that Python's package installer is available.

4. Create a Workshop Folder

```
mkdir -p ~/csscm2026
cd ~/csscm2026
pwd
```

Keeping workshop files in one folder makes them easier to find.

5. Create a Virtual Environment (Recommended)

```
python3 -m venv .venv
source .venv/bin/activate
```

When activated, the terminal prompt usually begins with (.venv). A virtual environment keeps workshop packages separate from the system Python.

6. Upgrade pip Inside the Environment

```
python -m pip install --upgrade pip
```

After activating the virtual environment, python and pip refer to the environment's copies.

7. Install NumPy

```
python -m pip install numpy
```

Wait until installation completes without an error.

8. Verify NumPy

```
python -c "import numpy as np; print(np.__version__)"
```

A NumPy version number confirms that the import works.

9. Create Your Python Script with nano

```
nano h2o2_analysis.py
```

The nano text editor opens inside the terminal.

10. Enter a Small Test Program

```
import numpy as np

values = np.array([1, 2, 3, 4])
print(values)
print(values.shape)
```

This is only an environment test; it is not an answer to the practical.

11. Save and Exit nano

```
Ctrl + O    Save (Write Out)
Enter       Confirm the filename
Ctrl + X    Exit nano
```

The ^ symbol shown by nano means the Ctrl key.

12. Run the Script

```
python h2o2_analysis.py
```

If you are not using a virtual environment, run `python3 h2o2_analysis.py` instead.

13. Edit and Run Again

```
nano h2o2_analysis.py
python h2o2_analysis.py
```

Programming is iterative: edit, save, run, read the output or error, then correct the program.

14. Leave the Virtual Environment

```
deactivate
```

Use this when you have finished the workshop session.

15. Return to the Environment Later

```
cd ~/csscm2026
source .venv/bin/activate
python h2o2_analysis.py
```

You do not need to reinstall NumPy each time. Reactivate the existing environment.

Common Beginner Problems

Problem / message	What to check
python3: command not found	Install Python 3 using the apt commands in Step 2.
No module named 'numpy'	Activate .venv, then install NumPy with: <code>python -m pip install numpy</code> .
externally-managed-environment	Do not force-install into system Python. Create and activate the virtual environment in Step 5.
SyntaxError	Check brackets, quotes, commas, colons and spelling near the indicated line.
IndentationError	Use consistent indentation, especially after for, if, elif, else and def.
NameError	Check whether the variable was created and whether its spelling/capitalisation matches.
File not found	Use <code>pwd</code> and <code>ls</code> to confirm you are in the folder containing the .py file.
Permission denied	For this workshop, run the script through Python rather than trying to execute the .py file directly.

Readiness Checklist

- Python 3 runs successfully.
- pip is available.
- The workshop folder exists.
- The virtual environment can be activated.
- NumPy imports successfully and prints a version number.
- You can create and save a .py file in nano.
- You can run the script and see output in the terminal.
- You know how to reopen the file, edit it and run it again.

Ready: Once every item above is checked, you are prepared for the CSSCM 2026 NumPy chemistry practical.